

Q1 (12 points)

Using careful measurements, you have determined subsequent ionization energies (in kJ/mol) for element X given below:

688.3(IE1)

1457.3(IE2)

2928.9(IE3)

4244.5(IE4)

17213.3(IE5)

20109.2 (IE6)

a. Which main group (in the periodic table) would you place element X in? Explain your reasoning.

b. Using the ionization energy data for element X given above, determine the longest wavelength (in nm) of a photon required to remove an electron from the outer "ns" orbital of the appropriate X cation.

ANSWER:

a. The biggest jump between ionization energies occurs from IE4 to IE5.

So, the cation must be at noble gas configuration when the electron is removed in IE5.

IE5 represents the following:



This means that the X^{4+} cation must have a noble gas configuration. So, the element X must belong to group 14 (or 4A).

b. The valence orbital configuration for group 14 elements is $ns^2 np^2$.

So, the first two ionization energies (IE1 and IE2) represent the removal of the np electrons.

IE3 represents the removal of the first ns electron.

$$IE3 = 2928.9 \text{ kJ/mol}$$

(convert kJ/mol to J)

$$IE3 (J) = 2928.9 \text{ kJ/mol} \cdot 1000 \text{ J/kJ} \cdot (1 \text{ mol} / (6.022 \times 10^{23})) = 4.864 \times 10^{-18} \text{ J}$$

$$\text{Energy} = hc / \lambda$$

$$\text{Solve for } \lambda = hc / E = (6.626 \times 10^{-34} \text{ J}\cdot\text{s} \cdot 2.998 \times 10^8 \text{ ms}^{-1}) / (4.864 \times 10^{-18}) = 40.8 \times 10^{-9} \text{ m} = 40.8 \text{ nm}$$

Q2 (12 points)

a. Two Chem110 students have decided to make a sensor using metal A, metal B, and metal C. They want to construct a sensor using a light source such that only two of the three metal surfaces eject electrons. The work function for metal A is 4.96 eV, for metal B is 5.97 eV, and for metal C is 3.82 eV. $1\text{eV} = 1.602 \times 10^{-19}\text{ J}$

Student 1 decides to use 275 nm wavelength light source, and Student 2 decides to use 225 nm wavelength light source. Which of the two students (1 or 2) is more likely to construct a successful sensor? Explain your answer and show calculations.

b. What will be the velocity (m/s; three significant figures) of an ejected electron of metal C (work function = 3.82 eV), if a light source of 400 kJ/mol is used?

ANSWER

a. Multiple different ways to get the answer.

Calculate longest wavelength required to observe photoelectric effect for A, B, and C.

Metal A longest wavelength required ~250 nm

Metal B longest wavelength required ~208 nm

Metal C longest wavelength required ~325 nm

So, the choice of wavelength which will excite electrons in two out of three metals must be <250 nm (metal A and C will eject electrons) but > 208 nm. Since wavelength is inversely proportional to energy, lesser than 250 nm wavelength will excite metal A and C electrons but not metal B (which require more energy)

Student 2 has the correct choice.

b.

Work function = 3.82 eV ----> convert to Joules

Work function (J) = $3.82 \times 1.602 \times 10^{-19}\text{ J} = 6.11976 \times 10^{-19}\text{ J}$

E of light source = 400 kJ/mol (Convert to J to get energy of one photon)

$E_{\text{photon}} = 400\text{ kJ/mol} \times 1000\text{ J/kJ} \times (1\text{ photon}/6.022 \times 10^{23}\text{ mol}) = 6.64231 \times 10^{-19}\text{ J}$

$E_{\text{photon}} = \text{Work function} + \text{KE}$

$6.64231 \times 10^{-19}\text{ J} = 6.11976 \times 10^{-19}\text{ J} + \text{KE}$

$\text{KE} = 5.2255 \times 10^{-19}\text{ J}$

$\text{KE} = \frac{1}{2}mu^2$

Mass of electron = $9.109 \times 10^{-31}\text{ kg}$

Solve for u

$u = 3.39 \times 10^5\text{ ms}^{-1}$

Q3 (10 points)

a.

Name the orbital, for Ca atom (Assume ground state electronic configuration)

(I) Orbital containing the electron that requires the maximum energy to be removed.

Name of the orbital:

(II) Orbital containing the electron that experiences the minimum effective nuclear charge.

Name of the orbital:

For the ground state Ca atom, draw the radial probability distribution for the above orbitals on one plot

b.

In a single atom:

(i) How many electrons can have the following two quantum numbers? $n = 5; m_l = -2$ (ii) How many electrons can have the following two quantum numbers? $n = 9; l = 2$

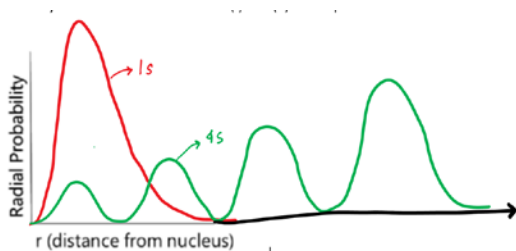
a. (I) Name the orbital containing the electron that requires the maximum energy to be removed.

1s orbital

(II) Name the orbital containing the electron that experiences the minimum effective nuclear charge.

4s orbital

(III) Radial Prob Distribution for I and II in one plot:

b. $n = 5; m_l = -2$ Following orbitals can have $n=5; m_l = -2$ $n=5; l=4; m_l = -2;$ $n=5; l=3; m_l = -2$ $n=5; l=2; m_l = -2$

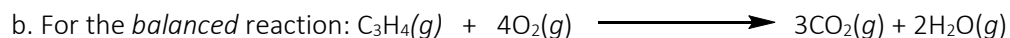
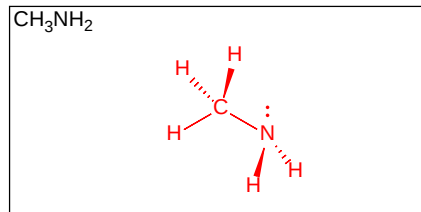
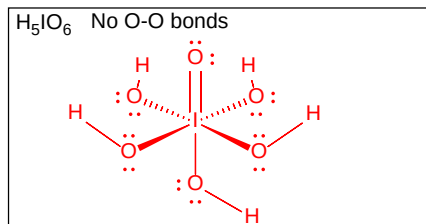
So, 6 total electrons

c. $n = 9; l = 2$

Can only be 9d orbital: 10 electrons

Q4:

For each molecule below, draw the most stable Lewis structure showing appropriate bond angles/shape and dashed/wedged bond where appropriate. Show lone pairs.



Calculate (in kJ) the standard enthalpy change (ΔH) for the combustion of 20.0 grams of propyne (C_3H_4). Give your answer to 3 significant figures.

Bond:	O=O	$\text{C}\equiv\text{C}$	C-C	C-H	O-H	C=O
Bond energy (kJ/mol):	347	839	346	413	467	799

Bonds broken: 4 mol C-H bond = $4 \times 413 = 1652$ kJ 1 mol C-C bond = $1 \times 346 = 346$ kJ

1 mol $\text{C}\equiv\text{C}$ bond = $1 \times 839 = 839$ kJ 4 mol O=O bond = $4 \times 347 = 1388$ kJ

Total = 4225 kJ

Bonds formed: 6 mol C=O bond = $6 \times 799 = 4794$ kJ

4 mol O-H bond = $4 \times 467 = 1868$ kJ

Total = 6662 kJ

So $\Delta H = 4225 - 6662 = -2437$ kJ

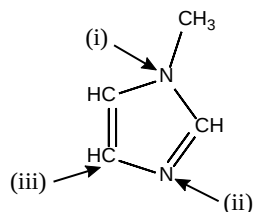
This is for 1 mole of propyne which is ~40.0 grams (molar mass for $\text{C}_3\text{H}_4 = 40.0$ g/mol), so for 20.0 grams it will be $-2437/2 = -1218.5$ kJ or 1220 kJ

c. Which of the following has a higher lattice energy? Explain your answer

KCl or CsI

Cs^+ is larger than K^+ and I^- is larger than Cl^- . Since both cation and anion are larger in CsI, the distance between the ions will be larger and hence the lattice energy will be lower.

5.



For the molecule and indicated atoms (denoted with an arrow, (i) to (iii)) answer the following questions. Consider the structure as given (no need to consider resonance structures). All atoms have formal charge of 0 (lone pairs are not shown). (Use valence bond theory and hybridization to describe the bonding.)

a) How many **total** sp^2 -s sigma bonds are in the given molecule

3 total

b) 1. Describe the orbitals involved in all bonds formed by N (i). Designate sigma/pi bonds.

N: 2 sp^3 - sp^2 sigma bonds ; 1 sp^3 - sp^3 sigma bond

2. What is the electron and molecular geometry at N (i)?

Electron Geometry: Tetrahedral ; Molecular Geometry: Trigonal Pyramidal

3. Circle the one value that best describes the C-N-C bond angle at N(i).

>90° to <109.5° 109.5° >109.5° to <120° 120° >120° to <180°

c) 1. Describe the orbitals involved in all bonds formed by N (ii). Designate sigma/pi bonds. (2 points)

N: 2 sp^2 - sp^2 sigma bonds ; 1 p-p pi bond

2. What is the molecular geometry at N (ii). (1 point)

Bent

3. Circle the one value that best describes the C-N-C bond angle at N (ii). (1 point)

>90° to <109.5° 109.5° >109.5° to <120° 120°

d) What is the electron and molecular geometry at (iii) (2 points)

Electron Geometry: Trigonal Planar ; Molecular Geometry: Trigonal Planar

Q6:

Arrange the following. Briefly explain your answer

1) In increasing order (smallest to largest) of bond angles

1. H_2S
2. XeF_2
3. $[SbF_6]^-$

2) In increasing order (smallest to largest) of size

1. Br^-
2. Po^{2-}
3. K^+
4. Sr^{2+}

3) In increasing order (smallest to largest) of bond length of CN bond in:

1. CH_3NH_2
2. HCN
3. CH_3CHNH

4) In increasing order (smallest to largest) of number of unpaired electrons:

1. Sc^+
2. Cr^+
3. As
4. Au^{2+}

1) $SbF_6^- < H_2S < XeF_2$ H_2S – Bent (bond angles < 109.5 – electron geometry is tetrahedral) XeF_2 : Linear (bond angle: 180) SbF_6^- : Octahedral (bond angle : 90)2) $Ca^{2+} < K^+ < Br^- < Po^{2-}$

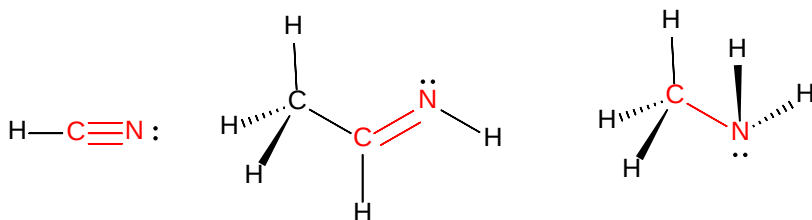
Po^{2-} is larger than At^- (they are isoelectronic and the higher negative charge of Po^{2-} means it is larger due to electron-electron repulsion)

Po^{2-} is thus larger than I^- (since I is in the same group as At, and At is further down the periodic table with valence electrons in larger n)

Ca^{2+} (this was changed from Sr^{2+}) and K^+ are isoelectronic – both are smaller than Br^- (since they are cations and are smaller than their isoelectronic anion Cl^-)

Ca^{2+} is smaller than K^+ as it has a higher positive charge and hence higher effective nuclear charge.

3) HCN (triple bond) < CH_3CHNH (double bond) < CH_3NH_2 (single bond)



4) $\text{Au}^{2+} < \text{Sc}^+ < \text{As} < \text{Cr}^+$

Sc^+ : 2 unpaired electrons

Cr^+ : 5 unpaired electrons

As: 3 unpaired electrons

Au^{2+} : 1 unpaired electron

Q7 (14 points)

a. In atomic spectra of H, a visible emission line is observed at 434 nm. Determine the level (n_{initial}) from which the transition occurred

b. Frodo is walking fast towards Mount Doom. What is the minimum uncertainty in his position, given that his velocity is $1000 \pm 2.0 \text{ mm} \cdot \text{s}^{-1}$? Frodo has a mass of 50 kg.

a.

H⁺: one electron species, so can use either Rydberg or Bohr's equation to solve for it.

Can use Rydberg or Bohr's equation.

Visible emission spectra so the n_{final} has to be smaller than n_{initial} (since energy is being emitted)

And since it is in the visible region, the n_{final} must be 2.

So, n_2 (in Rydberg) = n_{initial} (Bohr's equation) is the unknown

and n_1 (in Rydberg) = n_{final} (Bohr's equation) = 2

Since it is emission spectra:

$$Z = 1$$

$$\frac{1}{\lambda} = R \left(\frac{Z^2}{n_1^2} - \frac{Z^2}{n_2^2} \right)$$

Solve for n_2

$$R = 1.096776 \times 10^7 \text{ m}^{-1}$$

$$n_1 = 2 \quad n_2 = ?; \quad \lambda = 434 \text{ nm} = 434 \times 10^{-9} \text{ m}$$

$$1/(434 \times 10^{-9} \text{ m}) = 1.096776 \times 10^7 \text{ m}^{-1} \{ (1^2/2^2) - (1/n_2^2) \}$$

$$2.304 \times 10^6 \text{ m}^{-1} = 1.096776 \times 10^7 \text{ m}^{-1} \{ (0.25 - (1/n_2^2)) \}$$

$$2.1 \times 10^{-1} = 0.25 - (1/n_2^2) \quad \text{----->} \quad 0.21 - 0.25 = - (1/n_2^2)$$

$$-0.04 = -(1/n_2^2) \quad \text{So,} \quad 0.04 = (1/n_2^2)$$

$$n_2^2 = 25 \quad \text{Solve, } n_2 = 5$$

b.

Use equation for uncertainty principle here:

$$\Delta x \cdot \Delta p \geq h / 4\pi$$

$$p = \text{mass} \times \text{velocity}$$

$$\Delta p = \text{mass} \times \Delta v = 50 \text{ kg} \times (2.0 \times 10^{-3}) \text{ ms}^{-1} = 0.1 \text{ kg ms}^{-1}$$

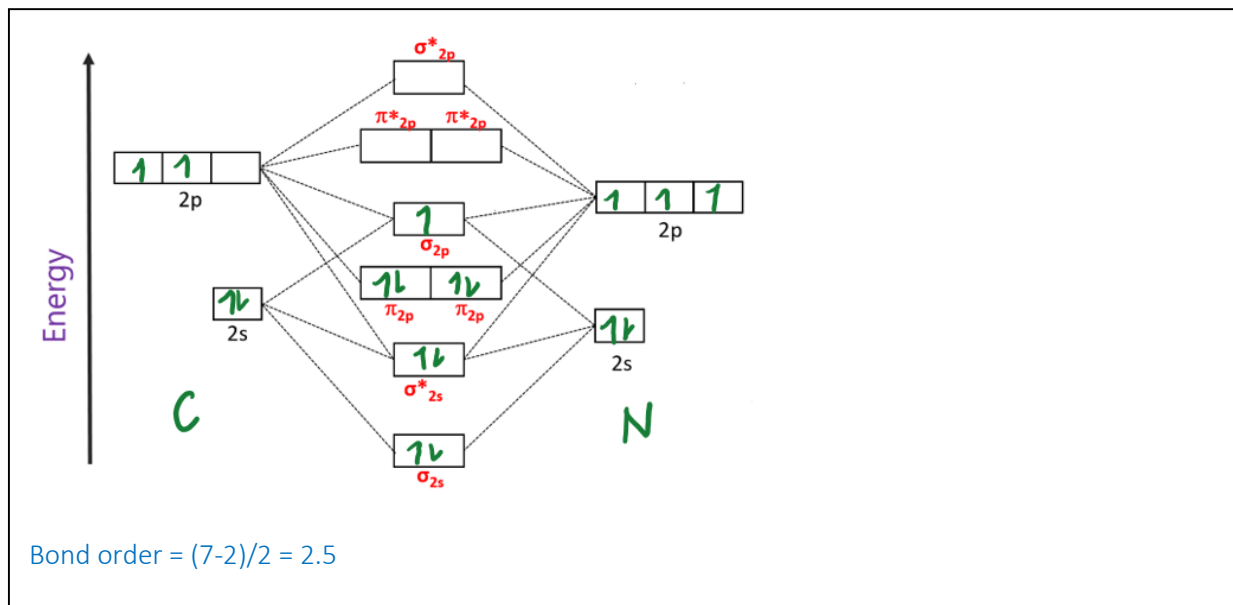
$$\Delta x \cdot (0.1 \text{ kg ms}^{-1}) \geq (6.626 \times 10^{-34}) \text{ Js} / (4\pi)$$

$$\Delta x (\text{minimum}) = \{(6.626 \times 10^{-34} \text{ kg m}^2\text{s}^{-1}) / (0.1 \cdot 4\pi \text{ kgms}^{-1})\}$$

$$\Delta x = 5.27 \times 10^{-34} \text{ m}$$

Question 8

- a. Draw the complete MO diagram (valence orbitals only) for the molecule CN. Make sure to label atomic and molecular orbitals. Label the energy axis. What is the bond order?



b.

Which of the following will require a higher energy to ionize (assume removal of easiest to remove electron)? Explain your answer

C, N, CN

CN will require the highest energy to ionize. Compare the MO diagram. Electron must be removed from the sigma bonding 2p orbital of CN which is lower in energy than the 2p atomic orbitals of C and N.